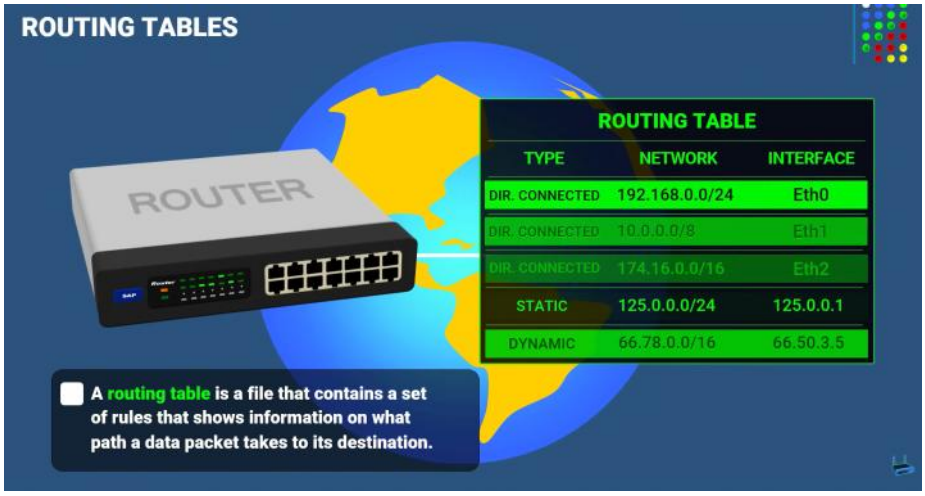
Routing Tables

Tuesday, April 21, 2026 3:50 PM

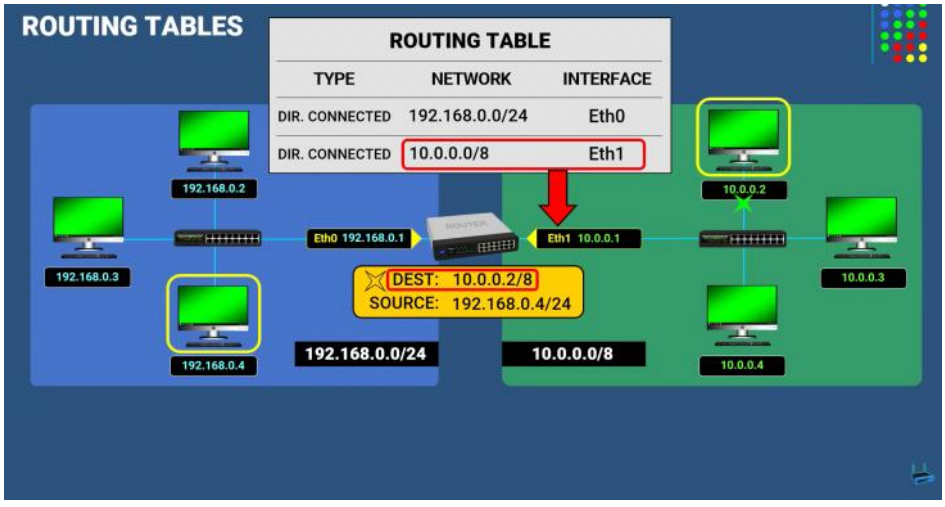
Routing Tables

A routing table is a digital file that contains a set of rules used by routers to determine the best path for forwarding data packets toward their destination, similar to how a google map provides directions for travel.

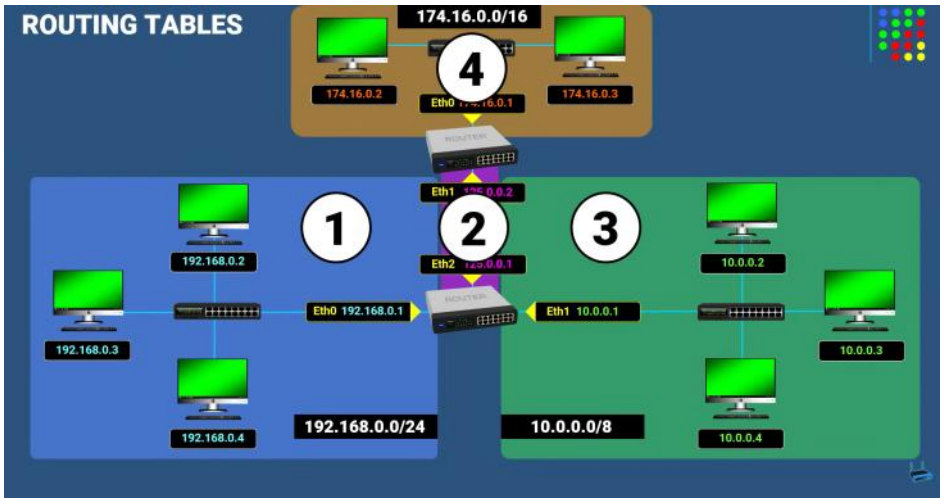


Routing tables are populated using three primary methods:

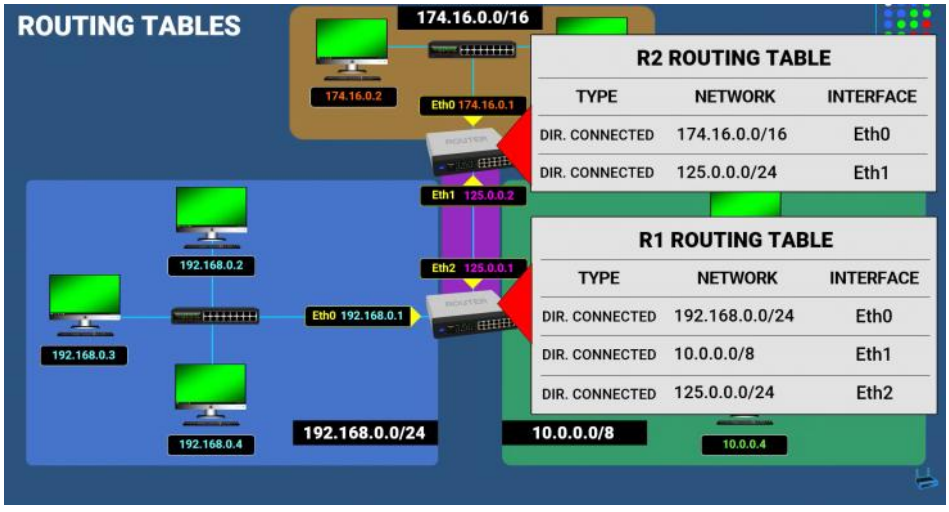
- Directly Connected: These routes are created automatically by the router whenever an interface is configured with an IP address and subnet mask. It informs the router about networks physically attached to its ports.

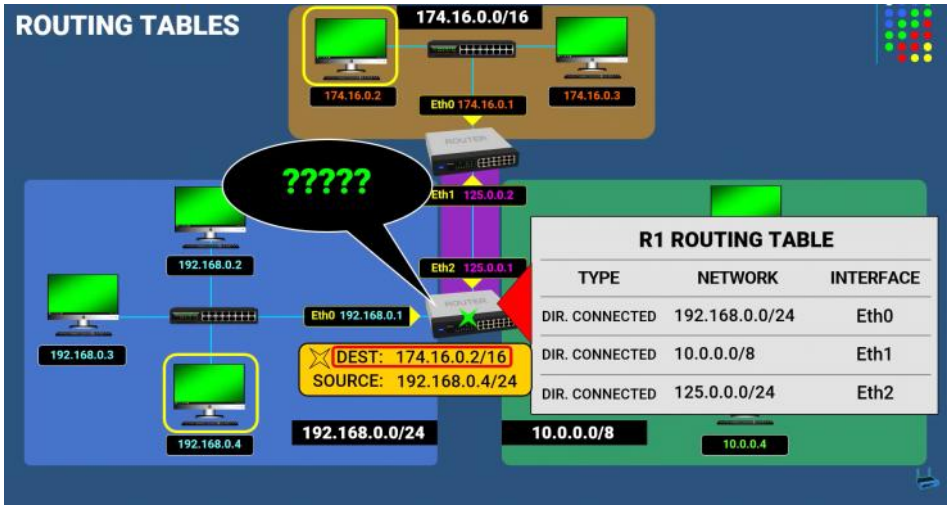


- Static: These routes are configured manually by a network administrator. They are necessary for the router to reach networks that are not physically connected to its own interfaces.

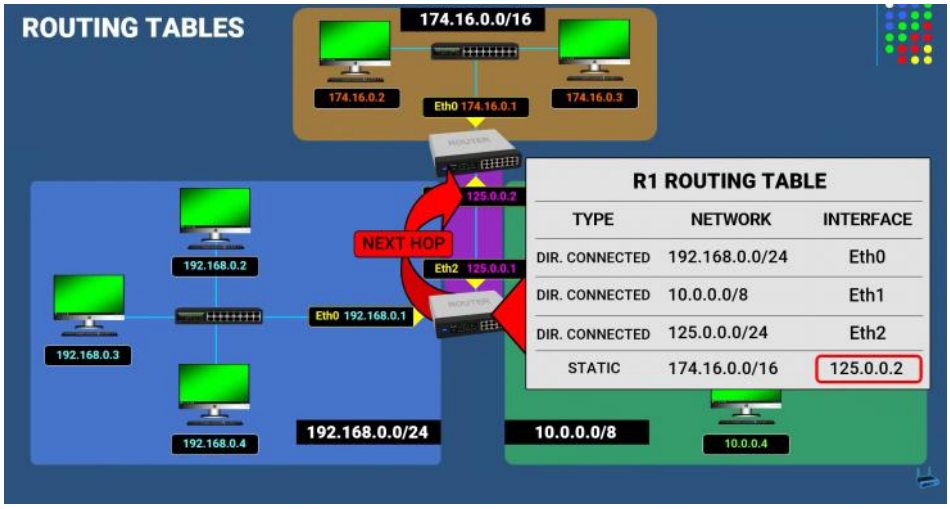


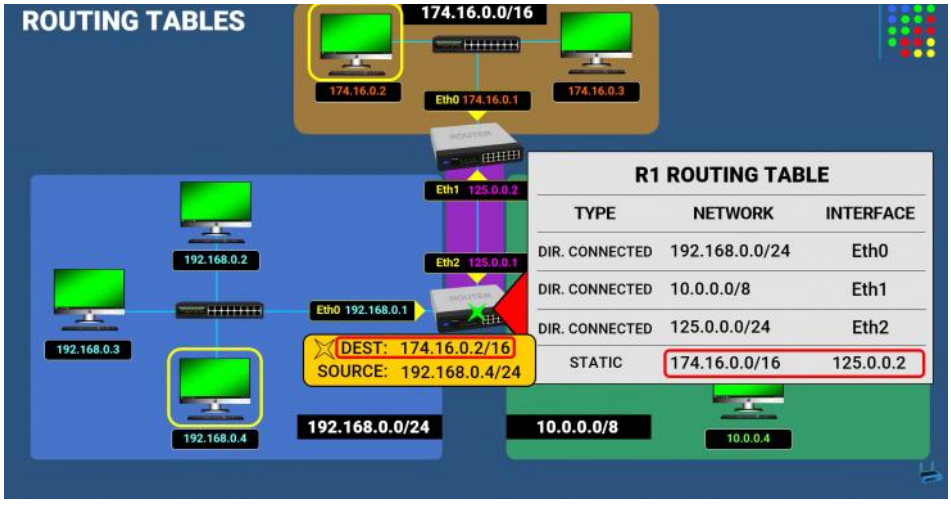
Here, there are 4 Networks including the 2nd one which is the network between two routers.

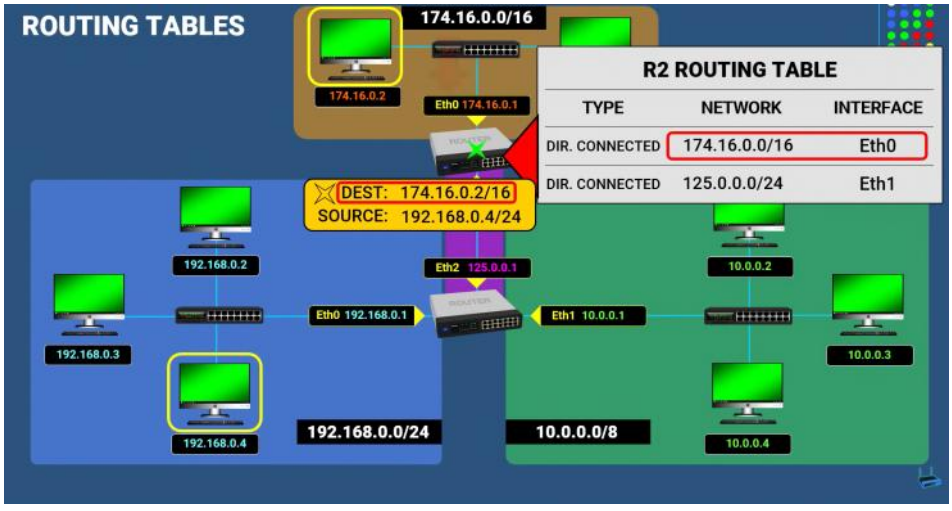


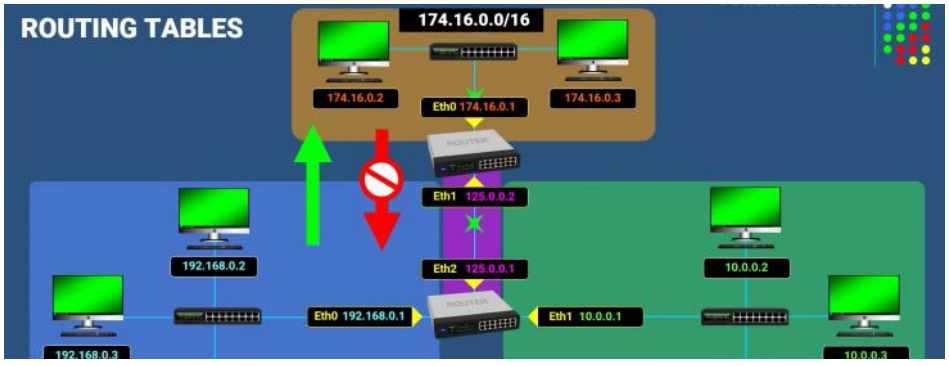


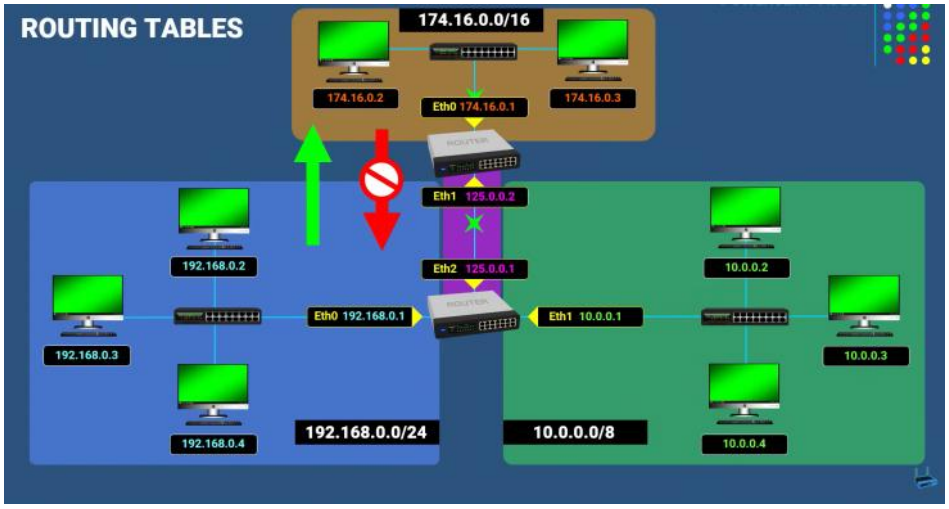
Here as you can see when the device .4 wants to communicate with .2 it checks for the destination IP on the router .1 but as it does not know that destination IP, it will just drop the packet so in order to avoid it we have use static routing to add a new route to the routing table manually.


ROUTING TABLES
174.16.0.0/16
174.16.0.2
Eth0 174.16.0.1
174.16.0.3
125.0.0.2
NEXT HOP
Eth2 125.0.0.1
192.168.0.2
Eth0 192.168.0.1
192.168.0.3
192.168.0.4
192.168.0.0/24
10.0.0.0/8
10.0.0.4
R1 ROUTING TABLE
TYPE NETWORK INTERFACE
DIR. CONNECTED 192.168.0.0/24 Eth0
DIR. CONNECTED 10.0.0.0/8 Eth1
DIR. CONNECTED 125.0.0.0/24 Eth2
STATIC 174.16.0.0/16 125.0.0.2

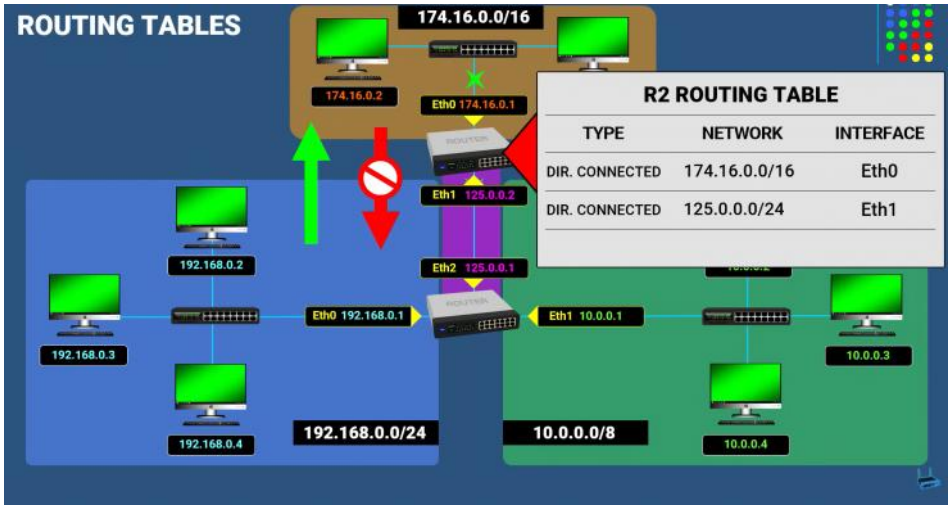

ROUTING TABLES
174.16.0.0/16
174.16.0.2
Eth0 174.16.0.1
174.16.0.3
Eth1 125.0.0.2
Eth2 125.0.0.1
192.168.0.2
Eth0 192.168.0.1
192.168.0.3
192.168.0.4
DEST: 174.16.0.2/16
SOURCE: 192.168.0.4/24
192.168.0.0/24
10.0.0.0/8
10.0.0.4
R1 ROUTING TABLE
TYPE NETWORK INTERFACE
DIR. CONNECTED 192.168.0.0/24 Eth0
DIR. CONNECTED 10.0.0.0/8 Eth1
DIR. CONNECTED 125.0.0.0/24 Eth2
STATIC 174.16.0.0/16 125.0.0.2

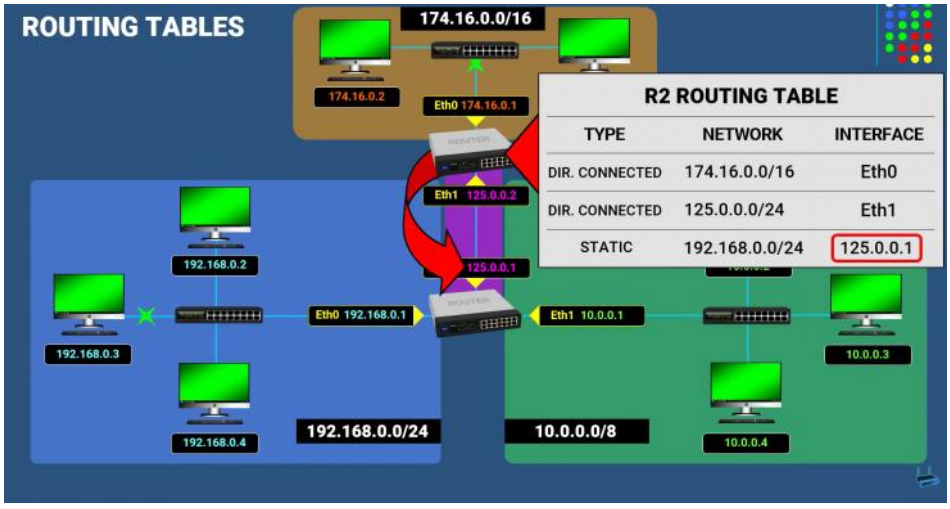

ROUTING TABLES
174.16.0.0/16
174.16.0.2
Eth0 174.16.0.1
DEST: 174.16.0.2/16
SOURCE: 192.168.0.4/24
192.168.0.2
Eth2 125.0.0.1
10.0.0.2
Eth0 192.168.0.1
Eth1 10.0.0.1
192.168.0.3
10.0.0.3
192.168.0.4
192.168.0.0/24
10.0.0.0/8
10.0.0.4
R2 ROUTING TABLE
TYPE NETWORK INTERFACE
DIR. CONNECTED 174.16.0.0/16 Eth0
DIR. CONNECTED 125.0.0.0/24 Eth1


ROUTING TABLES
174.16.0.0/16
174.16.0.2
Eth0 174.16.0.1
174.16.0.3
Eth1 125.0.0.2
192.168.0.2
Eth2 125.0.0.1
10.0.0.2
Eth0 192.168.0.1
Eth1 10.0.0.1
192.168.0.3
10.0.0.3

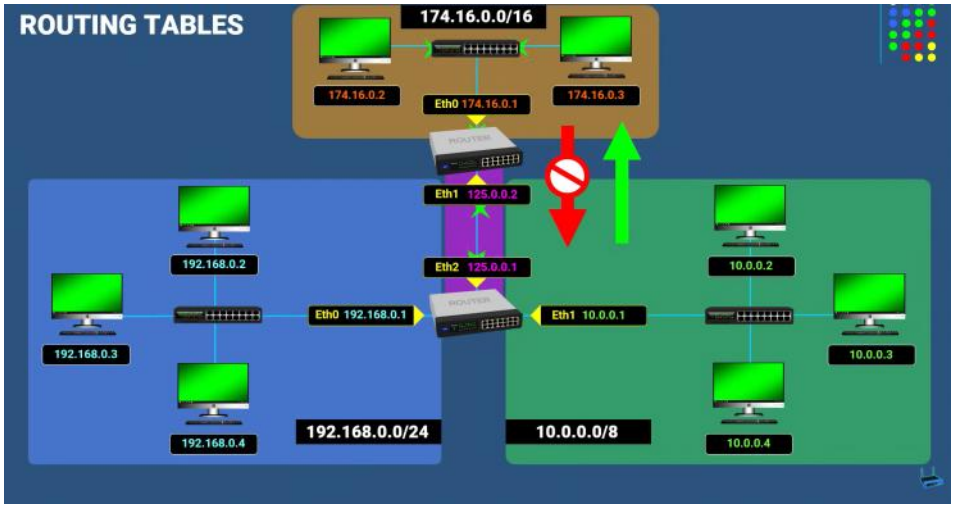


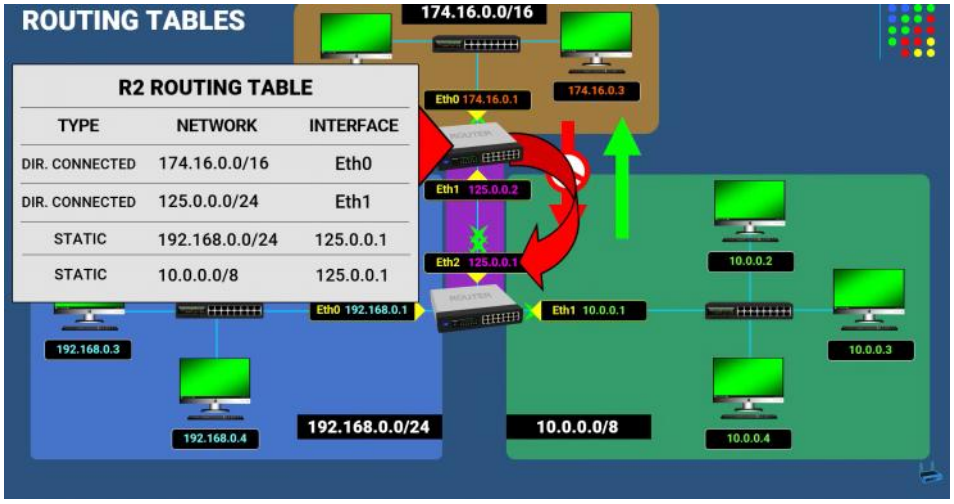
The only problem is that it will only communicate in one way so in order to make two way communication, we need to configure it in the other way also





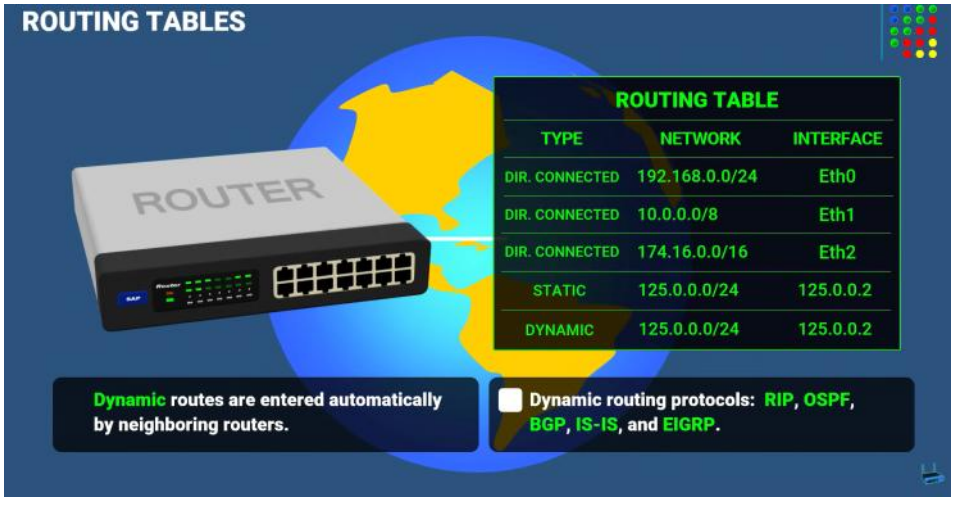
Also we have to do for the green network too as it can communicate with orange one but cannot communicate from orange to green as it is not configured so we have to do that in order for the network to communicate each other.

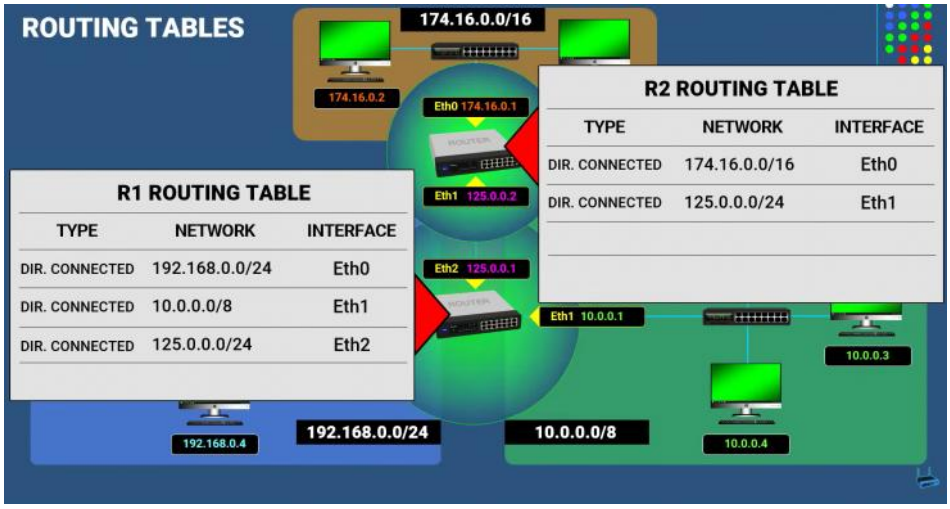


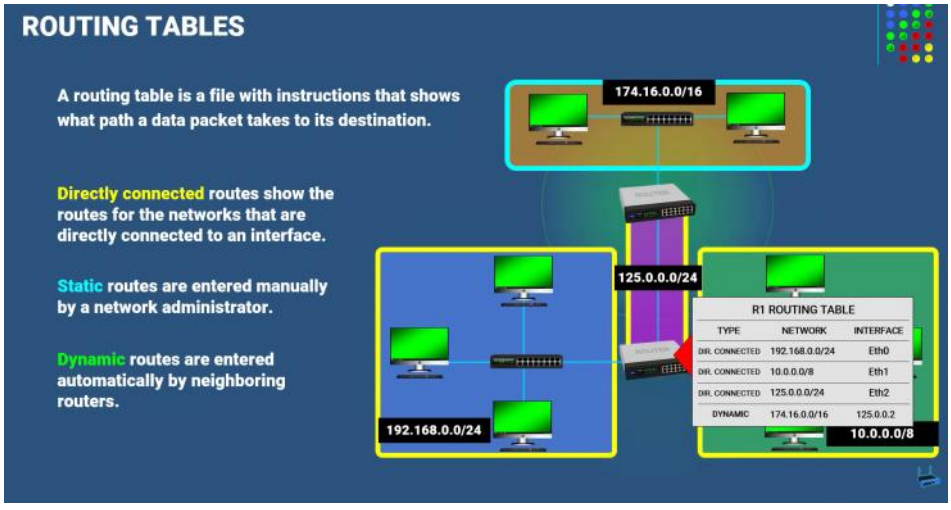


Now all the devices can communicate with each other on this network

- Dynamic: These routes are populated automatically by neighboring routers. Routers use dynamic routing protocols (such as RIP, OSPF, BGP, ISIS, and EIGRP) to communicate with each other, share their respective routing tables, and keep the network topology updated.




ROUTING TABLES
174.16.0.0/16
174.16.0.2
Eth0 174.16.0.1
Eth1 125.0.0.2
Eth2 125.0.0.1
Eth1 10.0.0.1
10.0.0.3
10.0.0.4
192.168.0.4
192.168.0.0/24
10.0.0.0/8
R2 ROUTING TABLE
TYPE NETWORK INTERFACE
DIR. CONNECTED 174.16.0.0/16 Eth0
DIR. CONNECTED 125.0.0.0/24 Eth1
R1 ROUTING TABLE
TYPE NETWORK INTERFACE
DIR. CONNECTED 192.168.0.0/24 Eth0
DIR. CONNECTED 10.0.0.0/8 Eth1
DIR. CONNECTED 125.0.0.0/24 Eth2


ROUTING TABLES
A routing table is a file with instructions that shows what path a data packet takes to its destination.
Directly connected routes show the routes for the networks that are directly connected to an interface.
Static routes are entered manually by a network administrator.
Dynamic routes are entered automatically by neighboring routers.
174.16.0.0/16
125.0.0.0/24
192.168.0.0/24
10.0.0.0/8
R1 ROUTING TABLE
TYPE NETWORK INTERFACE
DIR. CONNECTED 192.168.0.0/24 Eth0
DIR. CONNECTED 10.0.0.0/8 Eth1
DIR. CONNECTED 125.0.0.0/24 Eth2
DYNAMIC 174.16.0.0/16 125.0.0.2